

289 Diplopia and strabismus after plaque brachytherapy in uveal melanoma patients. Sara Munkwitz, Michelle Falcone, Kara Cavuoto, Basil Williams, Chloe Shields

Introduction: Iodine-125 plaque brachytherapy is a safe and effective treatment for uveal melanomas; however, sensorimotor complications can occur. This study evaluated the characteristics of strabismus and diplopia post-plaque treatment for uveal melanoma.

Methods: Single-center, retrospective study of 375 patients who underwent Iodine-125 plaque brachytherapy for uveal melanoma treatment between May 1, 2014, and December 1, 2022. Data on demographics, procedure details, and complications were analyzed.

Results: A total of 375 patients (mean age 64 ± 14 years; 50.4% female) with uveal melanoma were treated with plaque brachytherapy (average follow-up time, 3.7 ± 2.3 years). A total of 321 muscles were manipulated, predominantly the lateral ($n = 135/321$) followed by the inferior rectus ($n = 56/321$), with most individuals undergoing muscle detachment and reattachment ($n = 309/321$). Post-operative diplopia occurred in 38.9% of patients. Spontaneous resolution was seen in 71.2% of patients, with most resolving by 6 months (58.2%). There was no significant difference in the presence of diplopia in those who had muscle manipulation compared to those who did not (39.0% vs 38.7%, $P = 0.097$). Type of strabismus was documented in 27 patients, with exotropia noted most often ($n = 12/27$). Twelve patients developed strabismus in the absence of diplopia, all of whom had visual acuities worse than 20/200 at time of documentation. In those who experienced diplopia, prism correction was required in 6.8% of cases, and only one patient underwent strabismus surgery.

Conclusion/Relevance: Iodine-125 plaque brachytherapy for uveal melanoma carries a moderate risk of diplopia, which typically resolves on its own. There is also a risk of developing strabismus due to sensory deprivation.

290 Validation of the postnatal growth and retinopathy of prematurity (G-ROP) screening criteria in a midsized Midwestern childrens hospital. Omi Pang, Nabiha Habib, Julie Conley, Lauren Fletcher-Morehouse, Brooke Geddie

Introduction: Retinopathy of Prematurity (ROP) is the leading cause of preventable childhood blindness in the United States. Current screening criteria follow the American Academy of Pediatrics (AAP) guidelines. However, the Postnatal Growth and Retinopathy (G-ROP) study has proposed newer criteria, showing 100% sensitivity for detecting type 1 ROP while reducing the number of infants requiring dilated retinal exams by 35.6%. This study aims to validate the G-ROP criteria in a midsized Midwestern childrens hospital identified through filtering tools of our electronic medical record (EMR).

Methods: We conducted a retrospective cohort study of 687 infants screened for ROP using the AAP guidelines between January 2018 and December 2022. We then applied the G-ROP criteria to the same cohort and evaluated the sensitivity and specificity of these guidelines for detecting type 1 ROP requiring treatment.

Results: Of the 687 infants, 448 (65.2%) met G-ROP criteria. Of these infants, 33 were found to have type 1 ROP and received treatment. The sensitivity of G-ROP for detecting type 1 ROP was 97.06%. 1 infant with type 1 ROP would have failed to be screened and receive treatment utilizing the G-ROP screening protocol.

Conclusion/Relevance: While G-ROP demonstrated high sensitivity, it did not match the 100% sensitivity from earlier studies. Specificity was low but still improved compared to AAP guidelines. The baby missed using G-ROP criteria was an outlier with twin-twin

transfusion syndrome. This study demonstrates that the G-ROP can be implemented utilizing EMR filtering tools, however expansion of the criteria may be necessary to screen infants with additional risk factors for ROP.

291 Cost effectiveness of refractive surgery for myopia in special needs children. Asha Parvathaneni, Fatema Ghasia, Margaret Reynolds, Lawrence Tychsen

Introduction: Myopia-induced visual impairment is common in children with neurodevelopmental disabilities (NDD). An estimated 31% are ill-suited to spectacle wear. The impaired vision may exacerbate the NDD, increasing care burdens and expense. Restoring vision can improve NDD behaviors and quality-of-life. Here we evaluate the cost-effectiveness of pediatric myopia-correcting surgical procedures using Quality-adjusted life year (QALY) calculations. QALY factors both gain in function and years of benefit.

Methods: Utility values (UV) correlated to UDVA improvement were generated using standard gamble (SG) and time trade-off (TTO) parent-proxy assays. Average fees for office visits, ophthalmic tests, laser use, implant, surgeon, anesthesiologist, facility charge, medications, follow-up visits and management of anticipated average rate of complications was summed to derive total procedure cost (PC). PC entailed analysis of 152 PRK, 116 Ophtec phakic IOL (pIOL), and 153 Visian ICL pIOL surgeries. Life years (LY) of benefit was calculated as average lifespan minus patient age at surgery. Cost per QALY was derived by dividing PC by the product of UV and LY.

Results: SG and TTO UV were 0.83 and 0.88, respectively. LY averaged 65.3 years. The cost/QALY for PRK was \$329; for the Ophtec pIOL, \$322; and for the Visian pIOL, \$321. For comparison, adult cataract surgery has a cost/QALY of \$2,143 for the first eye. Adult vitrectomy for proliferative retinopathy has a cost/QALY \$66,182. The expense of pediatric myopic PRK and pIOL surgery is 16%, and 6% respectively the cost/QALY of adult cataract surgery.

Conclusion/Relevance: Refractive surgery for NDD children is cost effective.

292 Pre-ordered dilation drops for retinopathy of prematurity screenings: a quality improvement initiative to reduce time spent in the pediatric ophthalmology clinic. Madeline Rocks, Lauren Hennein, Angela Moll, Mansoor Movaghar, Shagun Bhatia, Preeti Bansal, Caitlin Carter, Henry OHalloran, Laurel Moyer

Introduction: The purpose of this quality improvement study was to reduce clinic time for premature patients presenting for retinopathy of prematurity (ROP) screening by pre-ordering dilation drops to be administered by the caregiver prior to clinic appointments. A major goal of our intervention was to reduce clinic time regardless of insurance type.

Methods: All patients seen in an urgent pediatric ophthalmology clinic at a large academic childrens hospital for ROP screening were included. The primary outcome was the median clinic time from check-in to check-out. Baseline data were recorded from 2/1/2023-4/30/2023 ($n = 154$). During the intervention period from 5/1/2023-5/31/2024, dilation drops were pre-ordered to be administered by the caregiver 90 minutes before the appointment ($n = 764$). Through a mail-order pharmacy, a discount was negotiated for patients whose insurance did not cover the drops. Significance was set at $P < 0.05$ for analysis of continuous variables with one-way ANOVA.